

b. Modular arithmetic

Modular arithmetic introduction

- Last time we discussed divisibility, now we'll talk about "equating" integers via their remainders when divided by some fixed integer
- Recall that $m \mid (a-b)$ doesn't imply $m \mid a$ and $m \mid b$, rather it means...
 - that their remainder is the same when divided by m , this motivates:
- **Def.** If $a, b \in \mathbb{Z}, m > 0$, then we say $a \cong b \text{ mod } m$ or $a \cong_m b$ if $m \mid (a - b) \Leftrightarrow a \text{ mod } m = b \text{ mod } m$

Properties of Mod function

- **Thm.** Mod is an **equivalence relation**

- **Pf.** 1. Reflexive:

- $\forall a, a \cong_m a \text{ } m|(a - a) = 0$

- 2. Symmetric:

- if $a \cong_m b$ then $m|(a - b) \Rightarrow m|(b - a) \Rightarrow b \cong_m a$

- 3. Transitive:

- If $a \cong_m b, b \cong_m c$ so $m|(a - b)$ and $m|(b - c)$ and so $m|((a - b) + (b - c)) = a - c$,
and so $a \cong_m c$.

Equivalence in terms of Mod function

- Consider equivalence classes defined under $\cong_m$
 - i.e., $\forall m > 0$ and $a \in \mathbb{Z}$ define $[a]_m = \{b \mid a \cong_m b\}$
- Let's define operations of these classes:
 - $[a]_m + [b]_m = [a + b]_m$ (where equality is by definition)
 - $[a]_m \cdot [b]_m = [a \cdot b]_m$
- Let's prove these operations are “well-defined”:
 - Will prove for addition, the multiplication case will be an (easy) exercise!
 - i.e., we need to show that, if $a_1 \cong_m a_2$ and $b_1 \cong_m b_2$ then $a_1 + b_1 \cong_m a_2 + b_2$
 - By definition we have $m \mid (a_1 - a_2)$ and $m \mid (b_1 - b_2) \Rightarrow m \mid ((a_1 + b_1) - (a_2 + b_2))$ QED

More properties of equiv. classes of mod

- This addition is also associative (and it “inherits this” from integer addition)
 - i.e., $\forall a, b, c$ and for $m > 0$, $[a]_m + ([b]_m + [c]_m)$
 - $= [a]_m + [b + c]_m = [a + (b + c)]_m = [a + b]_m + [c]_m = ([a + b]_m) + [c]_m$
- To recap, there’s an important “meta” phenomenon going on here:
 - We start with a structure, say the integers, $\mathbb{Z}$
 - Then we describe an equivalence relation on $\mathbb{Z}$, such as mod, and define equivalence classes using this equivalence relation
 - Then we notice that certain functions defined on $\mathbb{Z}$ are also well-defined on the equivalence class.
 - Then we can create a new structure named “ $\mathbb{Z}/\cong_m$ ” that consists of the equivalence classes together with operations that respect those classes, say $+$, $\cdot$
- In this case, we define “the integers *mod m*” to be the structure $\mathbb{Z}_m = \mathbb{Z}/\cong_m$ together with operations $+$, $\cdot$
- However, there are **important differences** between the integers mod m and the integers themselves:
 - for instance in the former there’s no meaningful ordering (i.e., no “ $<$ ” or “ $>$ ” operators)

Relatively prime numbers, and Mult. Inverses

- **Def.** Two integers a, b are *relatively prime* (or co-prime), if $\gcd(a, b) = 1$.
- **Def.** Integer a has a *multiplicative inverse* mod m if $\exists b, b \cdot a \cong_m 1$
- **Thm.** If $m > 0$ and a are relatively prime, then a has a multiplicative inverse *mod* m
- **Pf.**
 - Recall from last time, we proved Bezout's identity. ("Suppose $d = \gcd(m, a)$. Then there exists n, b so that $d = n \cdot m + b \cdot a$ ")
 - So now, using this identity, there exists integers n, b so that $n \cdot m + b \cdot a = 1$
 - Then computing equivalence classes mod m , (aka "residue classes") and the fact that addition, multiplication translates over these classes:
$$\begin{aligned} [1]_m &= [n \cdot m + b \cdot a]_m = [n]_m \cdot [m]_m + [b]_m \cdot [a]_m \\ &= [n]_m \cdot [0]_m + [b]_m \cdot [a]_m = [0]_m + [b]_m \cdot [a]_m = [b \cdot a]_m \text{ QED} \end{aligned}$$

Uniqueness

- **Thm.** If a and m are relatively prime then the mult. inverse of $a \bmod m$ is unique
- **Pf.** Suppose, for contradiction, that both b and c are mult. inverses of a
(**note:** equalities in what follows are shorthand for equivalence mod m)
 - Then: $b = b \cdot 1 = b \cdot (c \cdot a)$ (why?)
 - $= (b \cdot c) \cdot a = (c \cdot b) \cdot a = c \cdot (b \cdot a) = c$ – because of how we defined b
- **Notation:** we use a^{-1} to denote the mult. inverse of a , when it exists

Second corollary of mult. Inverse existence

- **Cor.2:** if p is prime, and a, b are integers so that $p|(a \cdot b)$ then $p|a$ or $p|b$.
- **Pf.** WLOG suppose p doesn't divide a (i.e., could have just as well considered b)
 - Then we know $\gcd(a, p) = 1$.
 - Now by Bezout's identity, there exist integers x and y so that $x \cdot a + y \cdot p = 1$
 - Multiply both sides on the right by b , so $b = x \cdot a \cdot b + y \cdot p \cdot b$
 - But now p divides the RHS (why?)
 - because it divides both terms of the sum
 - So it must divide the LHS= b . QED
- Can also prove a similar result:
- **Thm. (*)** If m_1, m_2 are relatively prime, and $m_1|a$ and $m_2|a$ then $m_1 \cdot m_2|a$
 - Notice that this is very similar, but we switched divisor and the dividend